

Square Numbers — Practice

Attempt every question on paper, showing your working. Marks are shown on the right. Answers are on the separate answer key — try first, then check.

Objective (1 mark each)

Q1 • MCQ**[1]**

Which of the following is the square of an even number? (a) 169 (b) 225 (c) 196 (d) 441

Q2 • MCQ**[1]**

How many natural numbers lie between 5^2 and 6^2 ? (a) 9 (b) 10 (c) 11 (d) 12

Q3 • MCQ**[1]**

Which of the following is **NOT** a perfect square? (a) 4 (b) 9 (c) 24 (d) 100

Q4 • FILL IN THE BLANK**[1]**

The units (ones) digit of the square of 162 is _____.

Q5 • TRUE / FALSE**[1]**

A number that ends in 7 can be a perfect square. (True / False)

Q6 • ASSERTION-REASON**[1]**

Assertion (A): 90000 is a perfect square.

Reason (R): A perfect square always ends in an even number of zeros.

Choose: (a) Both A and R true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7

[2]

If a number ends with 3 zeros, how many zeros will its square have at the end? Explain in one line.

Q8

[2]

Without calculating, state which of these are surely **not** perfect squares and give the reason for each: 1057, 23453, 7928, 222000.

Q9

[2]

Write the perfect squares that lie between 90 and 150.

Long answer (3 marks each)

Q10

[3]

Show that 36 is a perfect square by the method of repeated subtraction of odd numbers, and state its square root.

Q11

[3]

Using the odd-number pattern, find 36^2 given that $35^2 = 1225$. Show each step.

Q12

[3]

A square garden has an area of 169 square metres. Find the length of one side, and explain how the units-digit rule helped you check your answer.

Total: 20 marks · Check your work against the answer key.